Airbnb Data Analysis

Price Distribution and Log-Normal Fit

To analyse Sydney's Airbnb listings, I filtered the dataset to include only listings where room_type is "Entire home/apt", and number_of_reviews is at least 15.

To see how prices are distributed, I fitted a Log-Normal model $\text{Log}-\text{Normal}(\mu, \sigma)$ by taking the natural log of the price values and calculating the sample estimators:

$$\widehat{\mu} = \frac{\sum_{i=1}^{n} \ln(X_i)}{n}, \qquad \widehat{\sigma} = \sqrt{\frac{\sum_{i=1}^{n} (\ln(X_i) - \widehat{\mu})^2}{n-1}}.$$

The Log-Normal model fits the price data well. Rental prices can never be negative and the vast majority of listings cluster around $200 to $400 per night. However, there is a long tail of expensive luxury properties stretching up past $1000 per night. A standard Normal distribution wouldn't work here because it is symmetrical and would predict negative prices. The red Log-Normal curve captures both the peak around $200 and the long right tail very accurately.

Review Scores

I created a single composite score called weighted_score using the weighted average formula across six review areas:

$$\begin{aligned}\text{Weighted Score} \\ &= 0.5S_{\text{rating}} + 0.1S_{\text{accuracy}} + 0.1S_{\text{clenliness}} + 0.1S_{\text{checkin}} + 0.1S_{\text{communication}} \\ &+ 0.1S_{\text{location}}\end{aligned}$$

Filtering down to listings in the Warringah suburb gives these summary statistics:

$$\text{Mean Score} = 4.844133$$

$$\text{Median Score} = 4.88$$

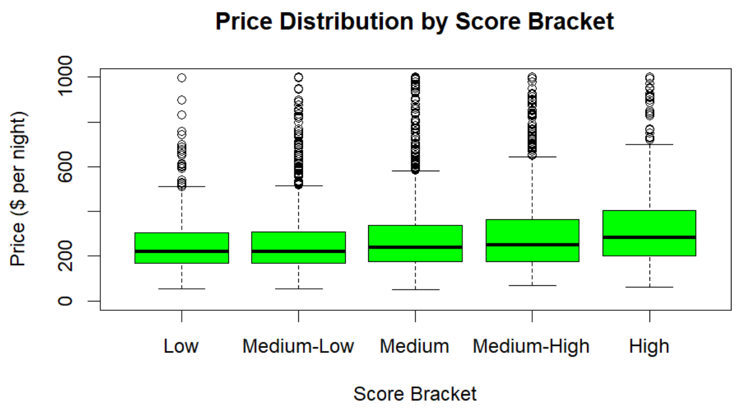
$$\text{Standard Deviation} = 0.13017$$

The median is higher than the mean. This happens because online review scores are heavily left-skewed as most guests leave 5-star ratings unless something goes wrong. The small standard deviation shows that nearly all listings in Warringah maintain high ratings, clustered tightly between 4.6 and 5.

I grouped all listings across Sydney into five rating tiers using the 10th, 35th, 65th, and 90th quantiles of weighted_score. Here is the average price for each bracket:

Score Bracket	Percentile Range	Mean Price per night
Low	Below 10^{th} percentile	$253.1343
Medium-Low	10^{th} to 35^{th} percentile	$257.9863
Medium	35^{th} to 65^{th} percentile	$283.6373
Medium-High	65^{th} to 90^{th} percentile	$299.3824
High	90^{th} percentile and above	$338.8396

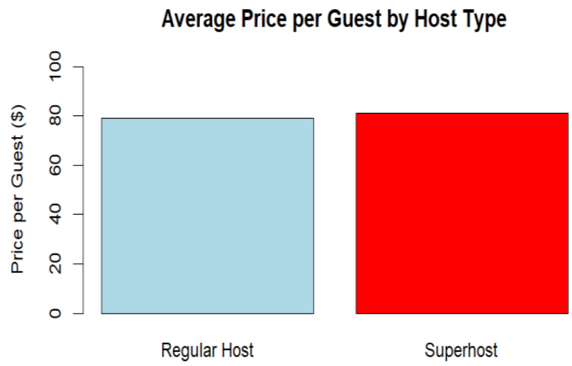
Price distribution by Score bracket Boxplot:



Average nightly prices increase steadily across brackets, moving from $253.13 in the low bracket up to $338.84 in the High bracket.

Moving from the Medium-High to High gives a big jump of $39.46 (+13.2%). Guests are willing to pay a noticeable premium for top-rated properties (top 10%).

Every boxplot shows tons of high price outliers (the open circles) shooting up to $1000. This means the while good reviews help hosts charge more, property size (accommodates), number of bedrooms, and ocean views still play much bigger role in setting the actual price than reviews scores alone.



A common idea is that becoming a Superhost lets you charge much higher rates. But comparing raw listing prices isn't fair because property sizes vary. To fix this, I divided price by capacity to look at the price per guest.

Superhost Mean Price per Guest:	$81.01
Regular Host Mean Price per Guest:	$79.05

Superhosts only charge about $1.96 more per guest per night (roughly 2.5% more). This shows an interesting host strategy: Superhosts don't use their badge to gouge prices. Instead, they keep their rates competitive to keep occupancy rates high, get favoured by Airbnb's search algorithm, and earn more total money through consistent bookings.